LECTURE 2: AGENT ARCHITECTURES

Artificial Intelligence II – Multi-Agent Systems

Introduction to Multi-Agent Systems

URV, Winter - Spring 2010

Outline of the talk

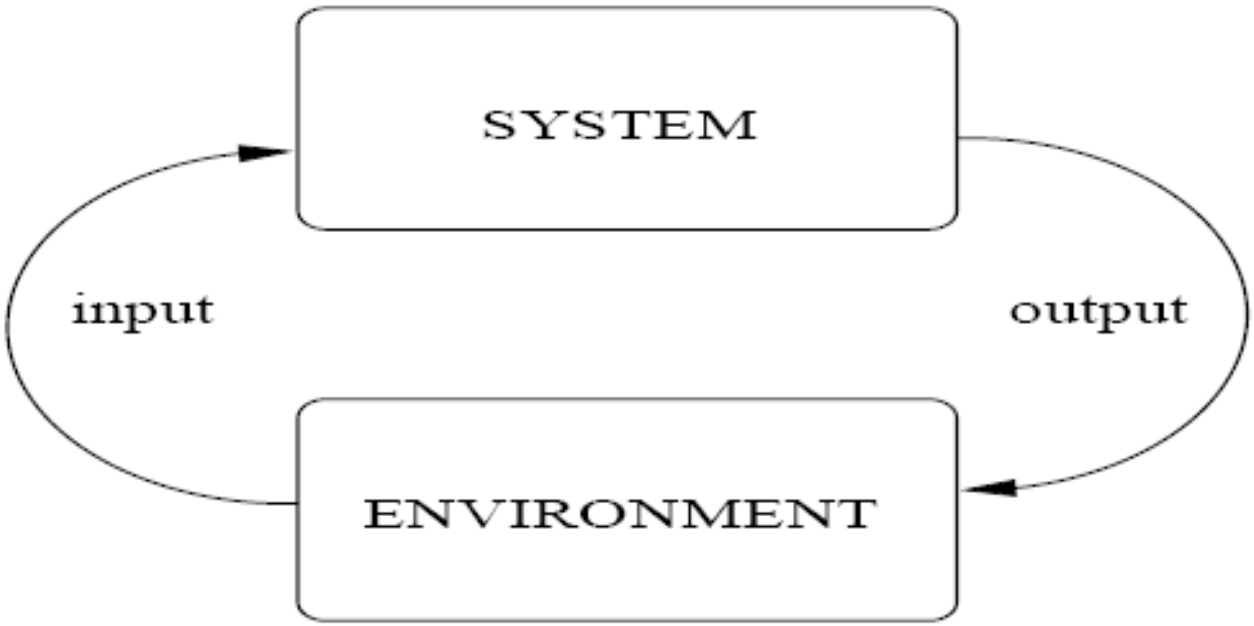
- Intelligent agent – reactivity
- Environments
- Agent architectures
  - Reactive
  - Deliberative
  - Hybrid

Agent definitions



- "An agent is anything that can be viewed as perceiving its environment through sensors and acting upon that environment through effectors."
- "Autonomous agents are computational systems that inhabit some complex dynamic environment, sense and act autonomously in this environment, and by doing so realize a set of goals or tasks for which they are designed."
- "An autonomous agent is a system situated within and a part of an environment that senses that environment and acts on it, over time, in pursuit of its own agenda and so as to affect what it senses in the future."

Basic abstract view of an agent



Reactivity

- An agent has to be able to react [adapt its behaviour] in an appropriate way to the dynamic changes in its "environment"
  - Other computational agents
  - Human agents/users
  - External information sources (e.g. sensors)
  - Physical objects (e.g. robots)
  - Internet
  - ...
- This is one of several properties that an intelligent agent should have

Kinds of environments (I)



- Accessible vs inaccessible
  - An accessible environment is one in which the agent can obtain complete, accurate, up-to-date information about the environment's state.
  - Most moderately complex environments (including, for example, the everyday physical world and the Internet) are inaccessible.
  - The more accessible an environment is, the simpler it is to build agents to operate in it.

Kinds of environments (II)

- Deterministic vs non-deterministic
  - A deterministic environment is one in which any action has a single guaranteed effect — there is no uncertainty about the state that will result from performing an action.
  - The physical world can to all intents and purposes be regarded as non-deterministic.
  - Non-deterministic environments present greater problems for the agent designer.

Kinds of environments (III)

- Episodic vs non-episodic
  - In an episodic environment, the performance of an agent is dependent on a number of discrete episodes, with no link between the performance of an agent in different scenarios.
  - Episodic environments are simpler from the agent developer's perspective because the agent can decide what action to perform based only on the current episode — it need not reason about the interactions between this and future episodes.

Kinds of environments (IV)

- Static vs dynamic

  - A static environment is one that can be assumed to remain unchanged except by the performance of actions by the agent.
  - A dynamic environment is one that has other processes operating on it, and which hence changes in ways beyond the agent's control.
  - The physical world is a highly dynamic environment.

Kinds of environments (V)

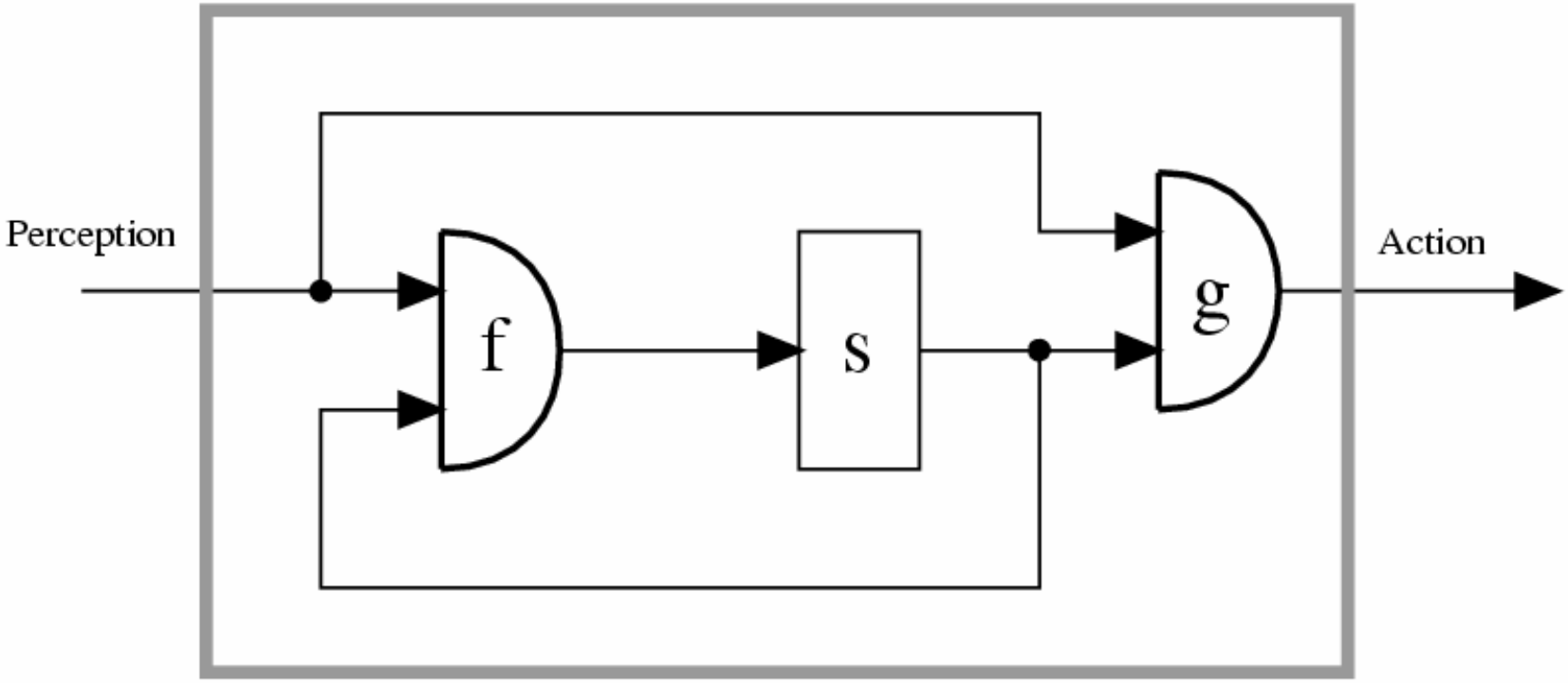
- Discrete vs continuous
  - An environment is discrete if there are a fixed, finite number of actions and percepts in it.
  - The real world is a continuous environment.

Practical exercise, step 1: think about the kind of environment

Agent architectures

- An architecture proposes a particular methodology for building an autonomous agent
  - How the construction of the agent can be decomposed into the construction of a set of component modules
  - How these modules should be made to interact
  - These two aspects define how the sensor data and the current internal state of the agent determine the actions (effector outputs) and future internal state of the agent

From perception to action



f = state update function
s = internal state
g = output function

Main kinds of agent architectures

- Reactive architectures
  - Focused on fast reactions/responses to changes detected in the environment
- Deliberative architectures (symbolic)
  - Focused on long-term planning of actions, centred on a set of basic goals
- Hybrid architectures
  - Combining a reactive side and a deliberative side

Reactive vs Deliberative: example

- Robot that has to reach a certain point
  - Reactive
    - Sensor in the front of the robot
    - Change movement right/left when sensor detects obstacle
      - Minimal computation based on current location and destination point
  - Deliberative
    - Explicit representation of the environment (map)
    - Planning procedure that finds the minimal route between the current position and the destination
      - High computational cost
      - Possible dynamic re-plannings needed

Reactive agents –basic ideas

- Reactive agents have
  - at most a very simple internal representation of the world,
  - but provide tight coupling of perception and action
- Behaviour-based paradigm
- Intelligence is a product of the interaction between an agent and its environment

Classic example: ant colony

- A single ant has very little intelligence, computing power or reasoning abilities
- The union of a set of ants and the interaction between them allows the formation of a highly complex, structured and efficient system.



Observation by Viviana Mascardi: as stressed during the class, an ant colony is an example of multiagent system, not of a single agent architecture: I disagree with the contents of this slide



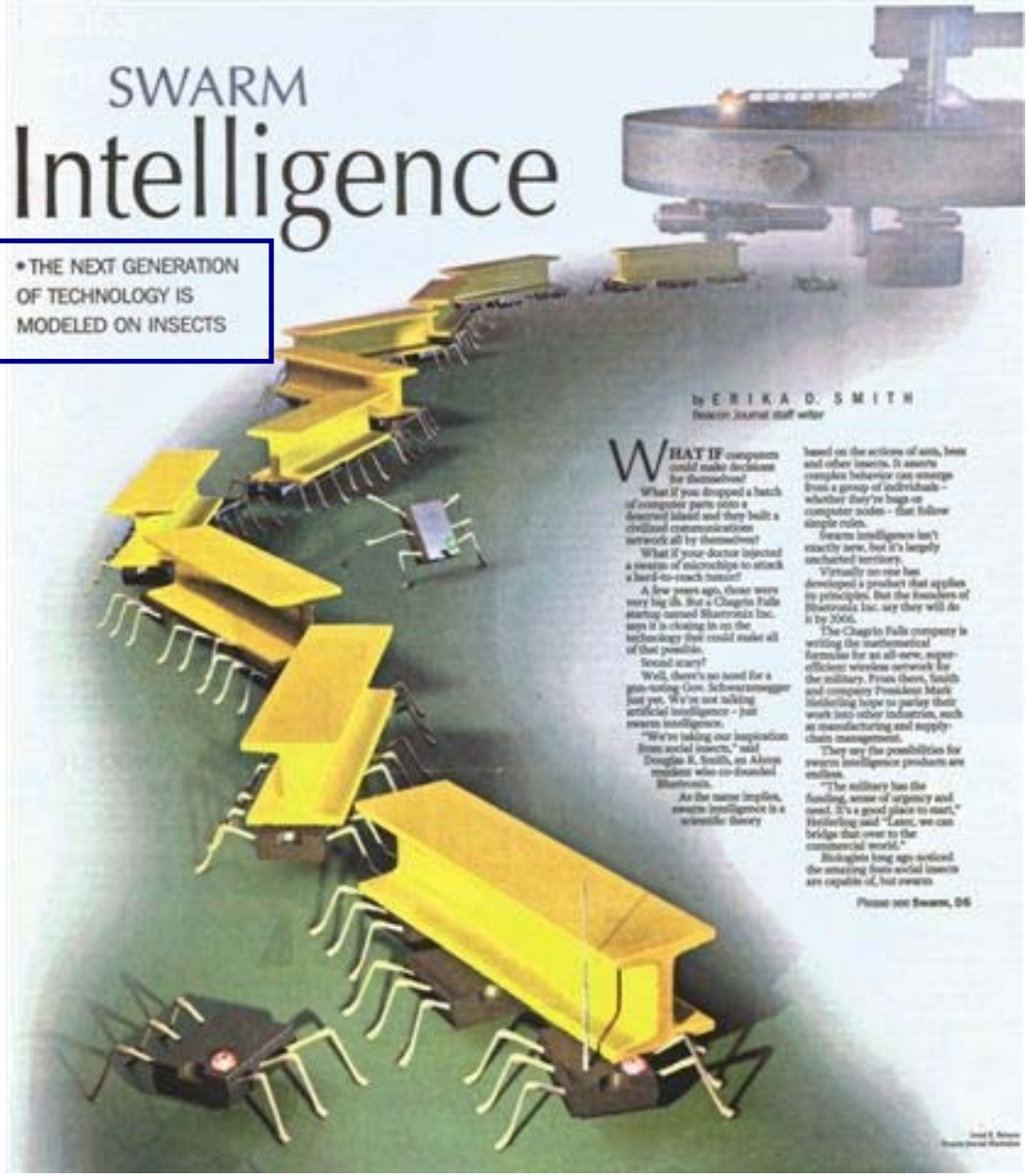






SWARM Intelligence

• THE NEXT GENERATION OF TECHNOLOGY IS MODELED ON INSECTS



by ERIKA D. SMITH
Beacon Journal staff writer

WHAT IF computers could make decisions for themselves?

What if you dropped a batch of computer parts onto a deserted island and they built a civilized communications network all by themselves?

What if your doctor injected a swarm of microchips to attack a hard-to-reach tumor?

A few years ago, those were very big ifs. But a Chagrin Falls startup named Bluetronix Inc. says it is closing in on the technology that could make all of that possible.

Sound scary?

Well, there's no need for a gun-toting Gov. Schwarzenegger just yet. We're not talking artificial intelligence – just swarm intelligence.

"We're taking our inspiration from social insects," said Douglas R. Smith, an Akron resident who co-founded Bluetronix.

As the name implies, swarm intelligence is a scientific theory based on the actions of ants, bees and other insects. It asserts complex behavior can emerge from a group of individuals – whether they're bugs or computer nodes – that follow simple rules.

Swarm intelligence isn't exactly new, but it's largely uncharted territory.

Virtually no one has developed a product that applies its principles. But the founders of Bluetronix Inc. say they will do it by 2006.

The Chagrin Falls company is writing the mathematical formulas for an all-new, super-efficient wireless network for the military. From there, Smith and company President Mark Heiderling hope to parlay their work into other industries, such as manufacturing and supply-chain management.

They say the possibilities for swarm intelligence products are endless.

"The military has the funding, sense of urgency and need. It's a good place to start," Heiderling said. "Later, we can bridge that over to the commercial world."

Biologists long ago noticed the amazing feats social insects are capable of, but swarm

Please see Swarm, D6

Main characteristics (I)

- Emergent functionality
  - Simple agents
  - Simple interaction
  - Complex behaviour patterns appear as a result of the dynamic interactions
  - The global behaviour of the system is not specified a priori
    - Dynamic movement of robots, depending on obstacles

Main characteristics (II)

- Task decomposition
  - Agents composed of autonomous modules
  - Each module manages a given task
    - Sensor, control, computations
  - Minimal, low-level communication between modules
  - There isn’t any world global model
  - There isn’t any “planning/controller agent”

Main characteristics (III)

- Raw data
  - Basic data from sensors
  - There isn't any complex symbolic management of data as in classical AI
    - Refusal of the *Hypothesis of the physic symbols system* [basic pillar of symbolic AI]
      - "Intelligent behaviour can only be obtained in symbol-processing systems"

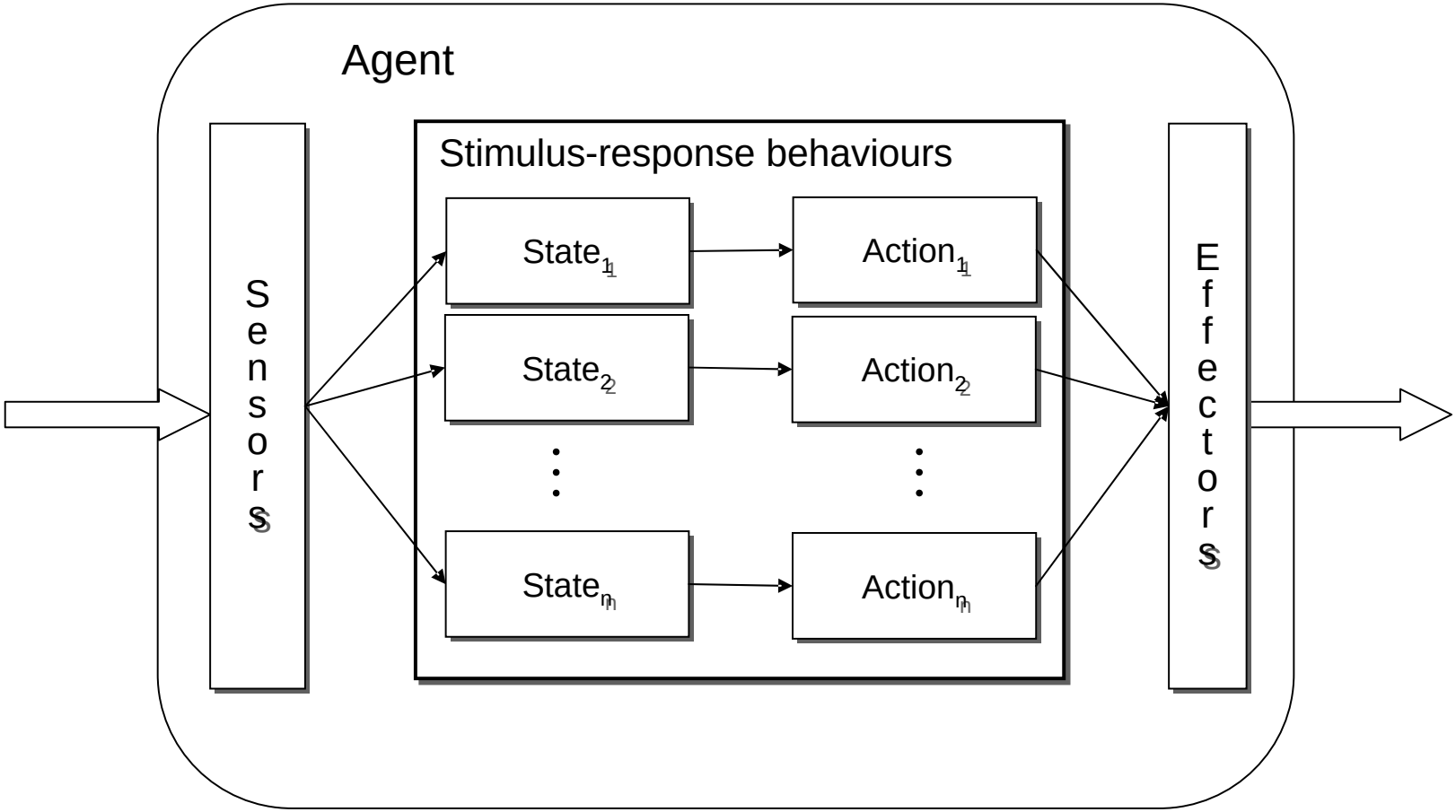
Basic concept

- Each behaviour continually maps perceptual input to action output
- Reactive behaviour: action rules: $S \rightarrow A$

  where S denotes the states of the environment, and A the primitive actions the agent is capable of performing.
- Example:

$$\text{action}(s) = \begin{cases} \textit{Heater off}, & \text{if temperature is OK in state } s \\ \textit{Heater on}, & \text{otherwise} \end{cases}$$

Basic schema of reactive arch.



Rodney Brooks



Director of the Computer Science and Artificial Intelligence Lab (MIT)

1997-2007

Brooks refutal of symbolic AI

- Brooks has put forward three theses:
  1. Intelligent behaviour can be generated *without* explicit representations of the kind that symbolic AI proposes
  2. Intelligent behaviour can be generated *without* explicit abstract reasoning of the kind that symbolic AI proposes
     - Reduced computation on sensor-like data
  3. Intelligence is an *emergent* property of certain complex systems

Brooks – key ideas (I)

- Situatedness: ‘Real’ intelligence is situated in the world
  - The world is its best model
  - The world is always up-to-date
  - A model is an abstraction, a simplification of the world, considering a particular set of characteristics and disregarding others

Brooks – key ideas (II)

- Embodiment: ‘Real’ intelligence requires a physical body, and cannot be found in disembodied systems such as theorem provers or expert systems
  - Physical robots
- Intelligence and emergence: ‘Intelligent’ behavior arises as a result of an agent’s interaction with its environment. Also, intelligence is ‘in the eye of the beholder’; it is not an innate, isolated property

Brooks – behaviours

- To illustrate his ideas, Brooks built some systems based on his *subsumption architecture*
- A subsumption architecture is a hierarchy of task-accomplishing *behaviours*
- Each behaviour is a rather simple rule-like structure
- Each behaviour ‘competes’ with others to exercise control over the agent, as different behaviours may be applicable at the same time

Behaviour layers

- Lower layers represent more primitive kinds of behaviour (such as avoiding obstacles)
- Higher layers represent more complex behaviours (e.g. identifying an object)
- Lower layers have precedence over layers further up the hierarchy
- The resulting systems are, in terms of the amount of computation they do, *extremely* simple
- Some of the robots do tasks that would be impressive if they were accomplished by symbolic AI systems

However....

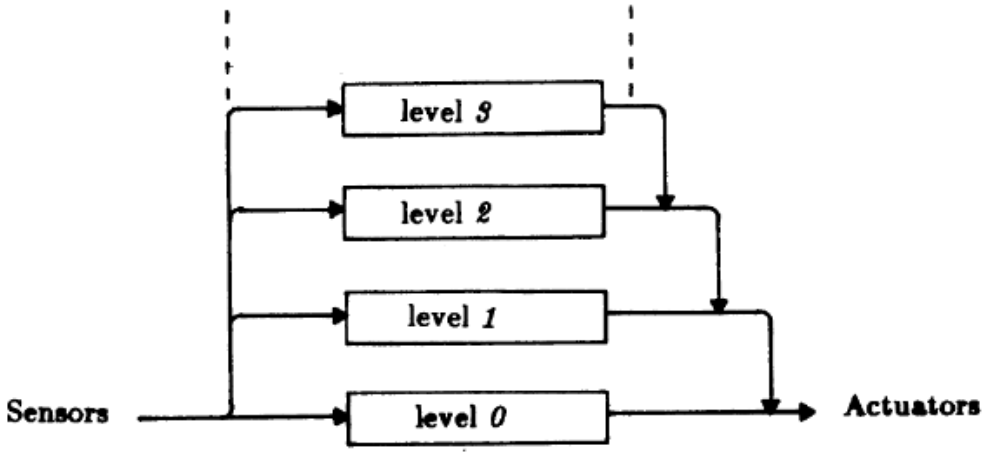


Figure 3. Control is layered with higher level layers subsuming the roles of lower level layers when they wish to take control. The system can be partioned at any level, and the layers below form a complete operational control system.

5. Reason about the world in terms of identifiable objects and perform tasks related to certain objects.
6. Formulate and execute plans which involve changing the state of the world in some desirable way.
7. Reason about the behavior of objects in the world and modify plans accordingly.

Notice that each level of competence includes as a subset each earlier level of competence. Since a level of competence defines a class of valid behaviors it can be seen that higher levels of competence provide additional constraints on that class.

Steels’Mars Explorer



- Steels’ Mars explorer system, using the subsumption architecture, achieves near-optimal cooperative performance in simulated ‘rock gathering on Mars’ domain:
  The objective is to explore a distant planet, and in particular, to collect sample of a precious rock. The location of the samples is not known in advance, but it is known that they tend to be clustered.

Steels' Mars Explorer Rules (I)

- For individual (non-cooperative) agents, the lowest-level behaviour, (and hence the behaviour with the highest "priority") is obstacle avoidance:

  if detect an obstacle *then* change direction (1)

- Any samples carried by agents are dropped back at the mother-ship:

  if carrying samples *and* at the base
  then drop samples (2)

- Agents carrying samples will return to the mother-ship:

  if carrying samples and *not* at the base
  then travel up gradient (3)

Steels' Mars Explorer Rules (II)

- Agents will collect samples they find:

  if detect a sample *then* pick sample up (4)

- An agent with "nothing better to do" will explore randomly:

  if true *then* move randomly (5)

Advantages of Reactive Agents

- Simplicity of individual agents
- Flexibility, adaptability
  - Ideal in very dynamic and unpredictable environments
- Computational tractability
  - Avoiding complex planning/reasoning procedures
  - Avoiding continuous model update
- Robustness against failure
  - No central planning component (e.g. ant colony)
- Elegance

I disagree: elegance is not a Scientific, engineering property. VM

I disagree: an ant colony is a MAS, not a single agent: a reactive architecture is as robust (or not robust) as any other modular software system. VM

Limitations of Reactive Agents (I)

- Agents without environment models must have sufficient information available from local environment
- If decisions are based on *local* environment, how can we take into account *non-local* information?
  - “Short-term” view
- No long-term planning capabilities
- Limited applicability
  - Games, simulations, basic robots (insects)

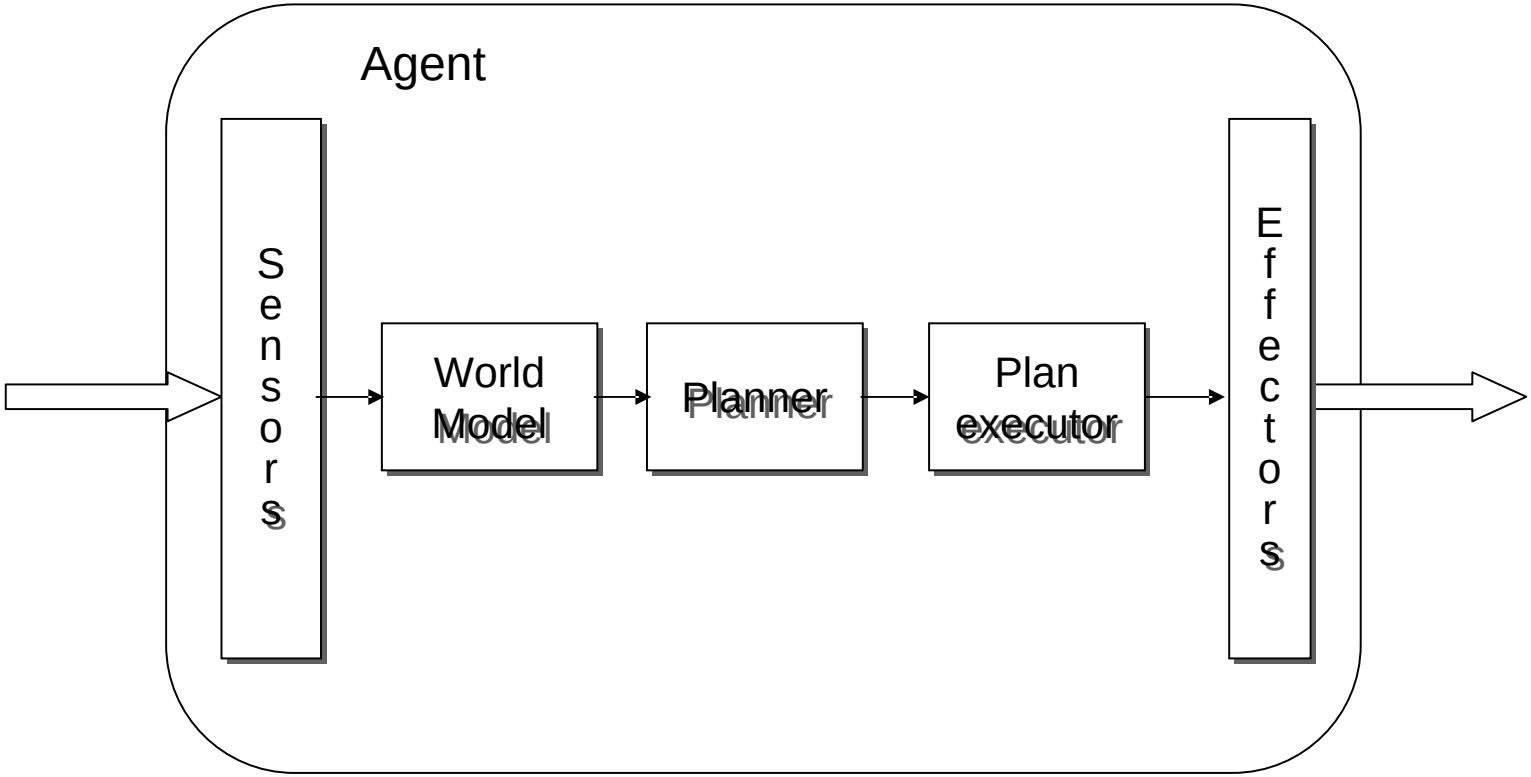
Limitations of Reactive Agents (II)

- Difficult to make reactive agents that learn
  - Dynamic evolution of rules?
- Since behaviour emerges from component interactions plus environment, it is hard to see how to *engineer* specific agents (no principled methodology exists)
- It is hard to engineer agents with large numbers of behaviours (dynamics of interactions become too complex to understand)

Deliberative agent architecture

- Explicit symbolic model of the world
- Decisions are made via logical reasoning, based on pattern matching and symbolic manipulation
- Sense-plan-act problem-solving paradigm of classical AI planning systems

Basic deliberative architecture



Problems in deliberative approach

- Dynamic world
  - Update symbolic world model
  - World changes while planning is being done
- Representation language
  - Expressive enough to be useful in any domain
  - Limited enough to be computationally tractable
- Classical planning => complete, optimal solutions
  - High computational cost
  - Sometimes a sub-optimal low-cost fast reaction can be effective

Hybrid Approaches

- Many researchers have argued that neither a completely deliberative nor a completely reactive approach are suitable for building agents
- They have suggested using *hybrid* systems, which attempt to marry classical and alternative approaches
- An obvious approach is to build an agent out of two (or more) subsystems:
  - a *deliberative* one, containing a symbolic world model, which develops plans and makes decisions in the way proposed by symbolic AI
  - a *reactive* one, which is capable of reacting quickly to events without complex reasoning

Hybrid agent architecture

